

The statistics of soaring flight

Displacement scaling across terrain, circuit types and wing classes

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OFFSITE
2026

EConophysics Lab

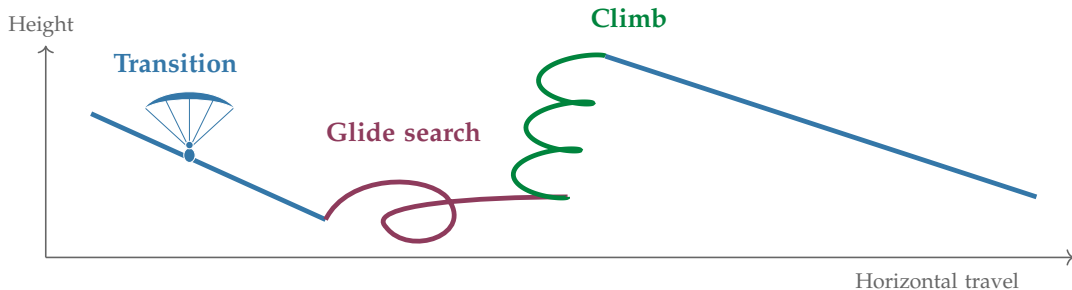
CFM

ILB Institut
Louis
Bachelier

Soaring is transport with an energy budget

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Height is spent to travel and recovered in rising air.

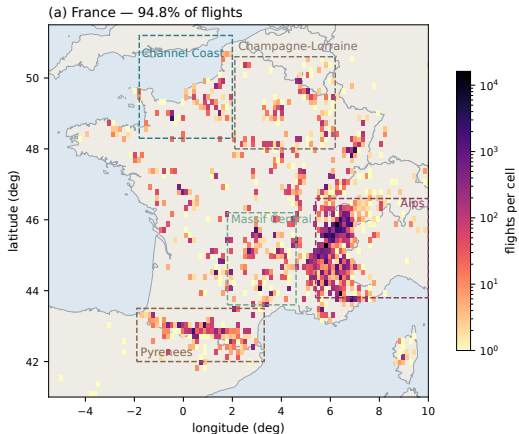


Wind, terrain and route choice act throughout the flight.

Which statistical laws emerge from this sequence of behaviours?

A large archive of free flight

3



FFVL distance competition

186,052

Paraglider tracklogs
1999–2000 to 2025–2026

6,716

Hang-glider tracklogs
2001–2002 to 2025–2026

Regional boxes

- Channel Coast
- Champagne-Lorraine
- Massif Central
- Alps
- Pyrenees

IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

France map: 148,081 paraglider and 5,894 hang-glider flights. Colour: flights per 0.15° cell.

Preprocessing: clean the recorded flight

4

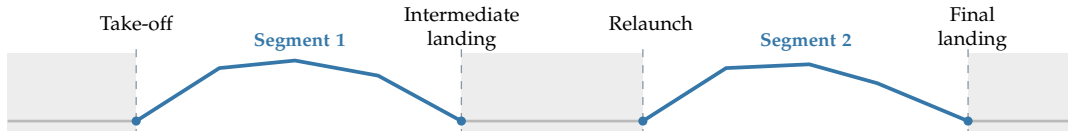
1 GNSS altitude

GNSS height $\longrightarrow$ Trajectory $z(t)$

Barometer $\longrightarrow$ Cleaning checks

GNSS: absolute height without weather drift.
Same high-frequency noise as the barometer.

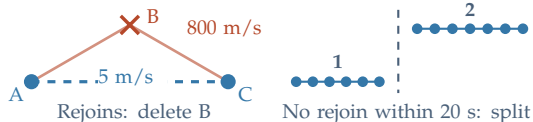
3 Airborne interval



Trim before take-off and after landing.
Stops are not recorded: we infer them from position and altitude.

2 Fix-level cleaning

One fix = (t, φ, λ, z) : time, latitude, longitude, height
Delete an unreachable fix only if the track rejoins without it; otherwise split the flight.



● Retained flight

■ Ground removed

4 Flight selection

Minimum flight thresholds

Time ≥ 40 min

1.13% excluded ($N = 191,108$)

Path ≥ 20 km

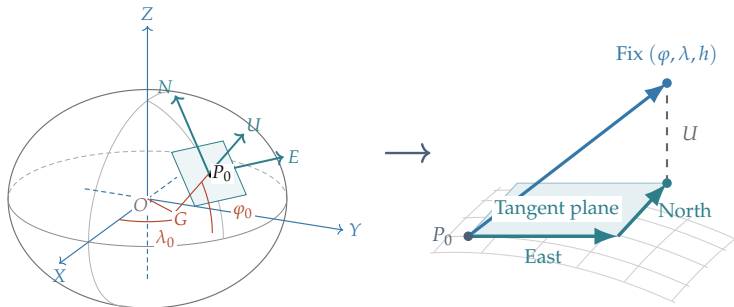
0.09% excluded ($N = 188,942$)

Altitude range ≥ 600 m

2.68% excluded ($N = 188,768$)

5 Coordinates

Latitude, longitude, height $\rightarrow$ Earth-centred position
 $\rightarrow$ local east-north distances in metres



Earth model: WGS84 ellipsoid, not a sphere (GPS reference)

P_0 : first trimmed fix; fixed for the flight. $z = h_{\text{GNSS}}$, not U .

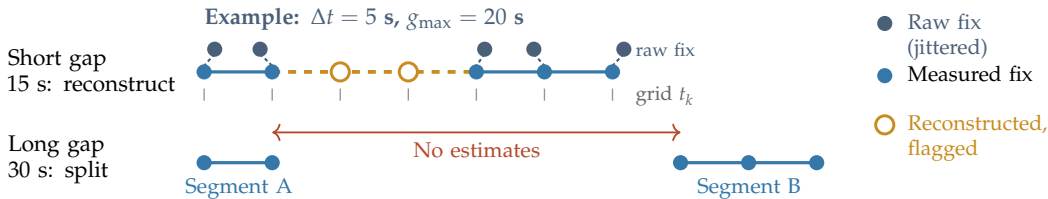
6 Time grid

Put each flight on an exact grid $t_k = t_0 + k\Delta t$ at its own native interval Δt .

- A fix within $\Delta t/2$ of t_k : use it.
- None that close: reconstruct t_k .

Reconstruct only a gap $g \leq g_{\max} = \min(10\Delta t, 20 \text{ s})$.

A longer gap splits the flight into segments.



Keep a segment only if it lasts $\geq 90 \text{ s}$ and $\leq 10\%$ of its grid is reconstructed.

No segment kept: flight dropped. 20,798 paragliders, 193 hang gliders: the largest loss.

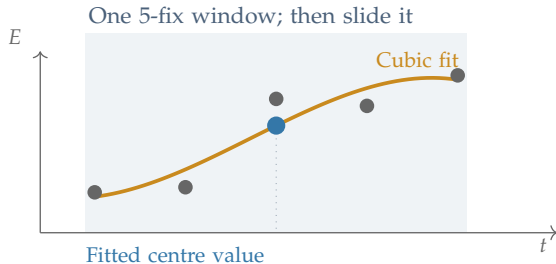
7 Savitzky-Golay fit

Fit a cubic to E , N , z in a moving window, within each segment.

Window: odd, ≥ 5 fixes.

Differentiate the polynomial analytically: no finite differences.

$$\frac{d^n \hat{x}}{dt^n}(t_i) = \left. \frac{d^n P_i}{du^n} \right|_{u=0}, \quad u = t - t_i$$



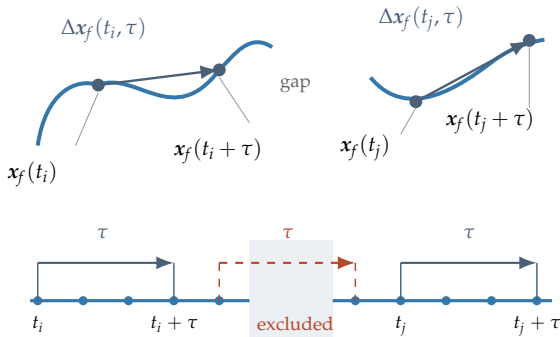
Retained flights after all seven stages (share of downloaded logs)

156,406 paragliders 84.1%

6,094 hang gliders 90.7%

Schematic; the cubic fit is exact for the synthetic fixes shown.

One flight f , two retained segments



- The solid arrows (Δx_f) lie in different segments, but each lies entirely within one: $i, j \in \mathcal{I}_f(\tau)$.
- The dashed arrow crosses the gap and is excluded.

For each lag τ :

- 1 Slide the origin t_i through each segment of f , one grid step at a time, while $t_i + \tau$ stays inside

$$\Delta x_f(t_i, \tau) = x_f(t_i + \tau) - x_f(t_i)$$
- 2 Pool these origins over all segments of f

$$\mathcal{I}_f(\tau) = \{i : t_i, t_i + \tau \text{ in the same segment}\}$$

- 3 Average the $n_f(\tau) = |\mathcal{I}_f(\tau)|$ squared displacements

$$m_f(\tau) = \frac{1}{|\mathcal{I}_f(\tau)|} \sum_{i \in \mathcal{I}_f(\tau)} |\Delta x_f(t_i, \tau)|^2$$

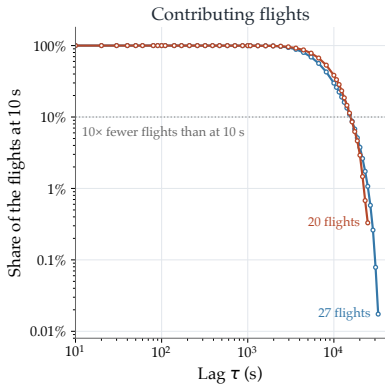
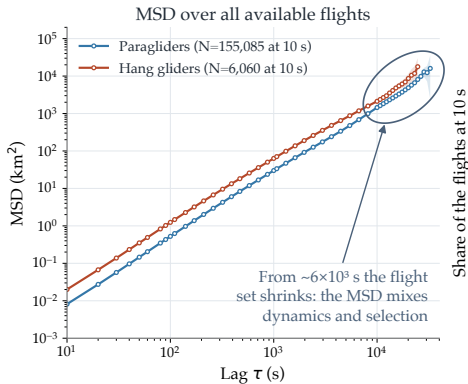
- 4 Average over the $N(\tau)$ flights with at least one origin, each with weight $1/N(\tau)$

$$M_2(\tau) = \frac{1}{N(\tau)} \sum_{f: |\mathcal{I}_f(\tau)| > 0} m_f(\tau)$$

Which flights contribute at each lag?

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All available flights: a flight enters at lag τ if one of its segments has $\tau \leq 0.8 T_s$ ($\geq 1/5$ of it left as origins for m_f).



Desiderata

- 1 The same flights at every lag
- 2 The longest possible lag range
- 3 $N \approx$ of the same order as at 10 s

At long lags only long flights remain, and their number falls steeply.

Any lag-dependent statistic then mixes dynamics with a changing set of flights.

Can we afford a fixed population?

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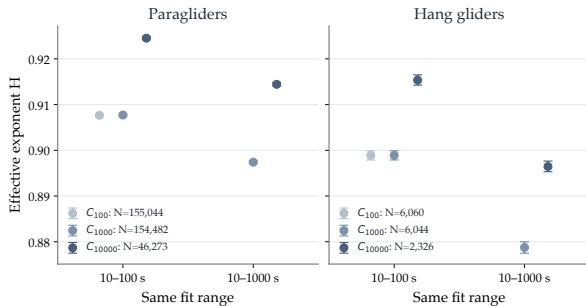
C_{10000} : select once for the largest lag

$\tau_{\max} = 10,000 \text{ s} \leq 0.8 T_s$: $T_s \geq 12,500 \text{ s}$
Only these segments enter, at every lag.

Advantages of C_{10000}

- 1 Same flights and segments, 10 to 10,000 s, with no loss of flights beyond $6 \times 10^3 \text{ s}$
- 2 Three decades of lag; at $\tau_{\max}$ each segment still offers at least 251 origins for m_f
- 3 Sample still large, same order as at 10 s:
46,273 paragliders (30%)
2,326 hang gliders (38%)

Cost of selecting longer flights



Nested cohorts, identical fit ranges, paired bootstrap draws.

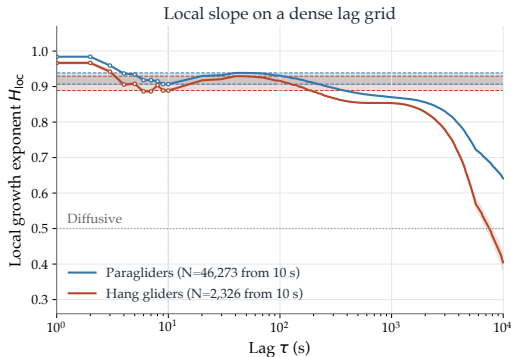
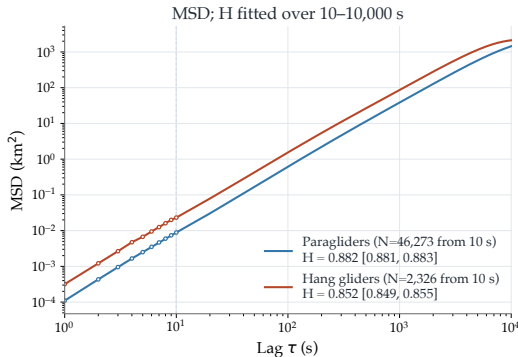
Long flights raise H by 0.016–0.018 on matched lag ranges; origins still decrease with lag.
About 2% in H buys the same flights at every lag over three decades.

Displacement growth from 1 s to 10,000 s

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C_{10000} : changing support below 10 s; the same flights at every lag from 10 to 10,000 s.

$M_2(\tau) \propto \tau^{2H}$: $H = \frac{1}{2}$ diffusion $H = 1$ ballistic motion



$$H_{PG} = 0.882 [0.881, 0.883] > H_{HG} = 0.852 [0.849, 0.855] \text{ over } 10\text{--}10,000 \text{ s}$$

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Which flight characteristics shape MSD and H ?

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Circuit type

Declared route

Open versus **closed**

Terrain / altitude

Proxy: launch altitude

Plains, **hills**, **low** and
high mountains

Experience

Proxy: wing class

Beginners: EN A/B/C
Experts: EN D/CCC

Weather also matters, but we have no data with sufficient spatial and temporal resolution.

Marginal comparison

Example: open vs closed flights
across all altitude classes.

Stratified comparison

Example: open vs closed flights
within each altitude class.

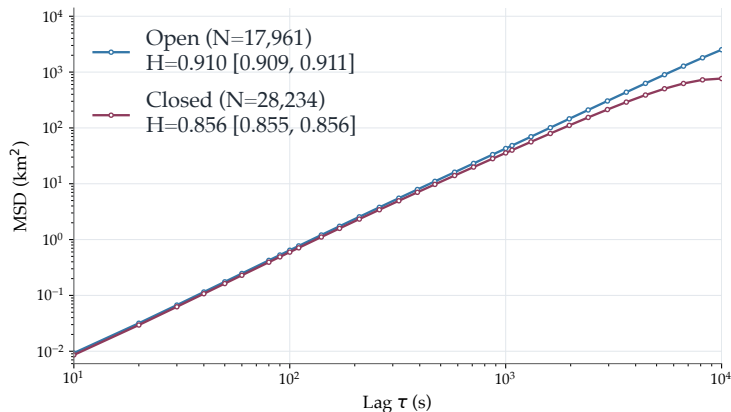
What stratification controls

The altitude class is shared; weather and equipment can still differ between groups.

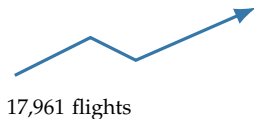
First compare groups overall, then within the same altitude class.

Open vs closed: how do MSD and H differ?

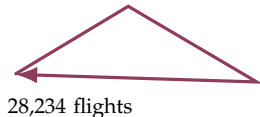
13



Open circuits



Closed circuits



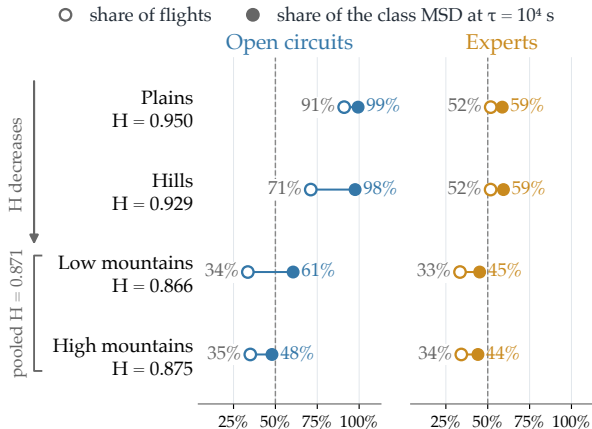
$$H_{\text{open}} - H_{\text{closed}} = 0.054 [0.053, 0.056] > 0 \quad (90\% \text{ paired interval})$$

Slower closed-route MSD growth is consistent with turns and returns.

78 unclassified flights excluded. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

Does the altitude ranking of H measure terrain alone?

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1 Different mix ○

Lowlands: more open circuits and more Experts, both with higher H .

2 Different weight ●

Group g , a circuit type or wing class, enters as flight share $\times$ MSD:

$$M_{2,a}(\tau) = \sum_g p_{g|a} M_{2,g,a}(\tau)$$

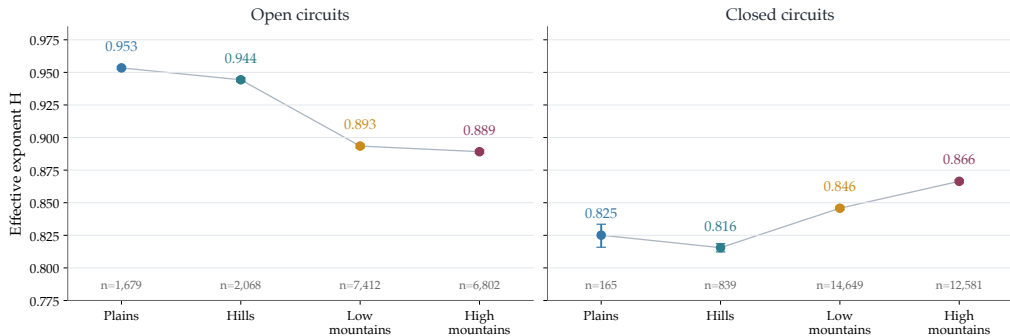
Low mountains: open flights are 34% of flights, 61% of the MSD at 10^4 s. Circuit type shifts the weights more than wing class.

Pooled H : Plains > Hills > mountains; the circuit and wing mix changes with altitude too.
An altitude-only comparison mixes terrain with circuit and wing composition.

Shares of all flights in each class; unclassified: circuit < 0.3%, wing up to 10%. Observed shares, no confidence bands.

Does the open–closed H gap change with altitude?

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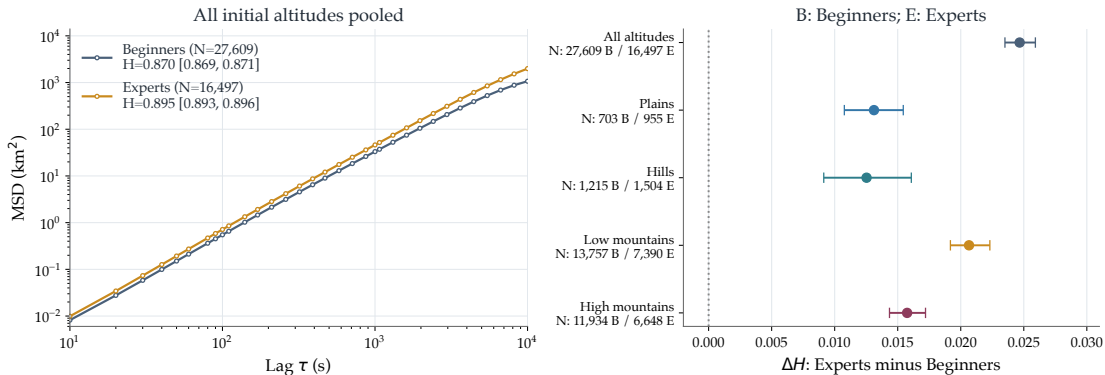
Open: lowland H is higher. In mountains, each transition must head for known lift, so the flight moves out less directly.

Closed: mountain H is higher. In plains, pilots can turn back sooner and more often; in mountains, even the way home must follow the lift.

Interaction: gap ≈ 0.13 in lowlands, 0.02 – 0.05 in mountains: the circuit effect depends on altitude.
Open > closed in every class; mountain orography constrains both, and the gap narrows.

Does the expert–beginner H gap change with altitude?

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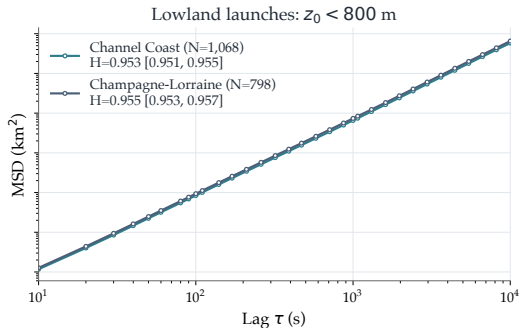
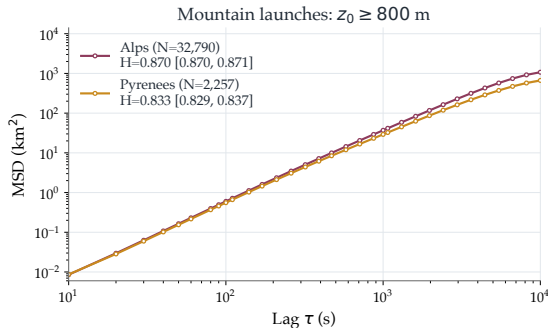
Higher H for experts in every class; the gap depends on altitude.

A constant speed advantage changes MSD amplitude, not H .

Wing class is a proxy. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

Within altitude groups, how does H differ by region?

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Mountain hypothesis

Slope-linked lift ties routes to ridge and valley geometry.

Lowland hypothesis

Roads and dark fields may supply broadly distributed thermal triggers.

ΔH : Alps–Pyrenees 0.037 [0.033, 0.041]; lowlands -0.002 [-0.005 , 0.001].

Similar lowland H ; Pyrenees may constrain routes more than the Alps.

How do climb locations change with height?

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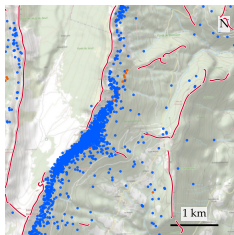
All dates and years pooled. Vilpellet climb crossings.

H2: Chartreuse, Alps

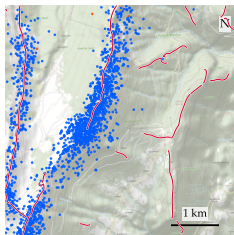
High mountains #2 / 15,840 flights

$z_{AGL} = 100 \text{ m}$

$z_{AGL} = 600 \text{ m}$



1,856 crossings / 1,261 flights
PG: 1,249 HG: 12



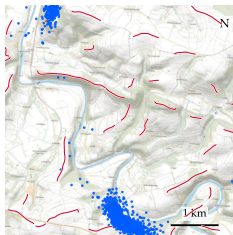
1,868 crossings / 1,452 flights
PG: 1,434 HG: 18

P1: Suisse Normande, Channel Coast

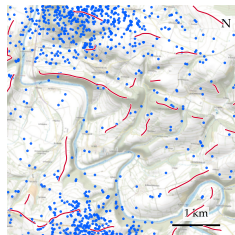
Plains #1 / 2,246 flights

$z_{AGL} = 100 \text{ m}$

$z_{AGL} = 600 \text{ m}$



1,441 crossings / 734 flights
PG: 733 HG: 1



901 crossings / 574 flights
PG: 574 HG: 0

100 to 600 m: P1 climbs spread out; H2 climbs stay near slopes and ridges.

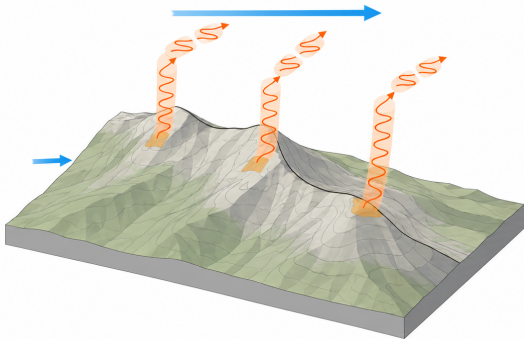
Observed lift use is less confined aloft in P1,
supporting more spatial freedom for route choices.

Why do lowland climb locations spread aloft?

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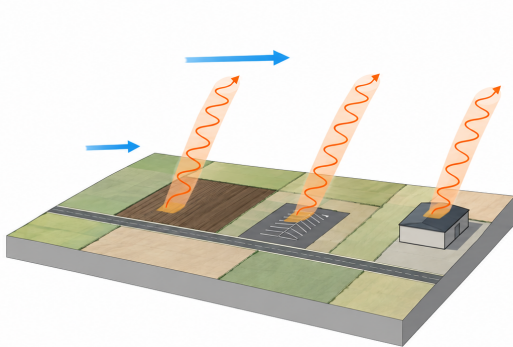
Mountains: sources on heated slopes

Possible shelter below crests; stronger shear aloft.



Lowlands: separated heated surfaces

Dry field, asphalt, warm roof; drift from the base.

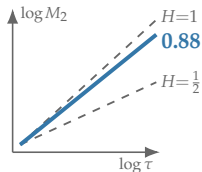


Distributed sources and variable wind could broaden lowland crossings.

Mountain slope sources and possible shelter can keep lift use closer to terrain corridors.

Orange: rising air; blue: wind. Illustrative profiles, not measurements. Sources: FAA, Ch. 9–10; SSA, Lift Sources.

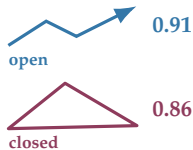
Scaling



Strongly superdiffusive

$H = 0.88$ for paragliders, 0.85 for hang gliders, fitted over $10\text{ s} - 10^4\text{ s}$ on the same flights at every lag.

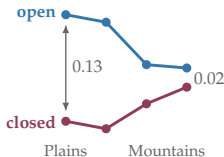
Route and wing



Open routes spread faster

Open exceeds closed in every altitude class ($+0.054$ pooled). Experts exceed beginners, by less ($+0.025$).

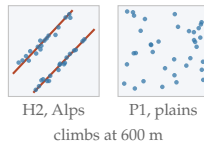
Terrain



Mountains narrow the gap

In mountains, open-route H falls and closed-route H rises. Altitude alone mixes terrain with route and wing mix.

Lift geography



Mountain lift hugs ridges

From 100 to 600 m, H_2 climbs keep to slopes and ridges; P_1 climbs spread out. Observed lift use; causes untested.

Strongly superdiffusive in every group; route and terrain interact, wing adds a small shift.

Working hypothesis: terrain constrains routes through where pilots can find lift.

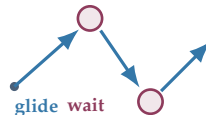
Segmentation



Revise Vilpellet's HMM

Tuned on 1 Hz tracks;
misses some of our climbs.
Next: continuous features
in physical time.

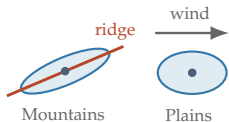
Stochastic model



Describe the observations

Candidate: glides alternate
with waits (search + climb),
built from measured glides,
turns and waiting times.

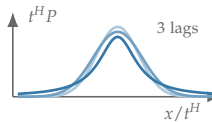
Anisotropy



Ridge axis or wind

Mountains: chain axis.
Plains: weaker, mean wind.
Preliminary observations.

Distribution scaling



Do distributions collapse?

Collapse and moment tests:
mixed so far. Power-law
MSD alone does not
establish self-similarity.

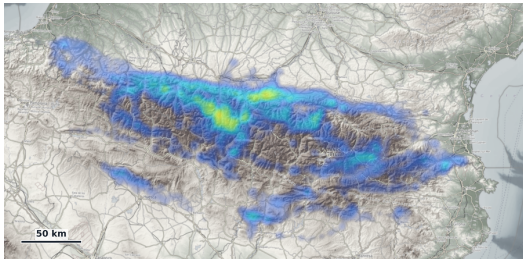
Thank you
for your attention!

Where do mountain climbs concentrate?

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Pyrenees: full chain

Atlantic to Mediterranean



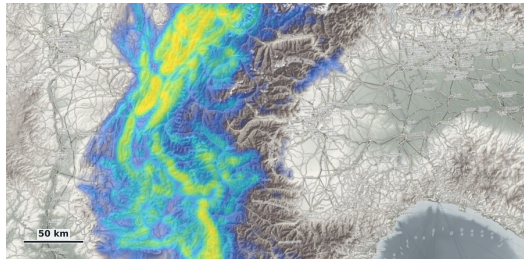
2,257 flights: open 698 (30.9%), closed 1,556 (68.9%)

$H = 0.833$

Climb crossings per km²

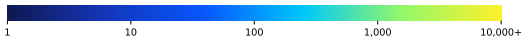
Western Alps

Matched 450 × 220 km area



32,790 flights: open 10,533 (32.1%), closed 22,212 (67.7%)

$H = 0.870$



Pyrenees: a narrow northern band. Alps: more branching climb areas.

Similar open shares make terrain guidance plausible, but the maps do not establish causation.

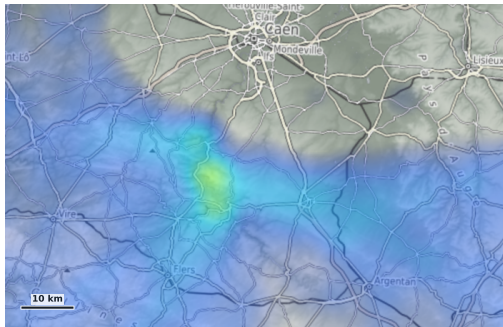
C10000 paragliders; crossings every 10 m, all dates pooled. Equal 450 × 220 km views and colour range. Map: © OSM contributors, SRTM, OpenTopoMap (CC-BY-SA).

How broadly are lowland climbs distributed?

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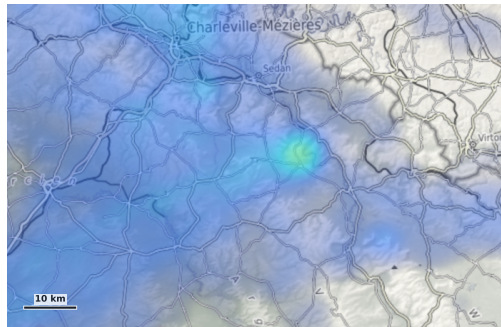
Channel Coast

1,068 flights, $H = 0.953$, 95.5% open



Champagne-Lorraine

798 flights, $H = 0.955$, 92.9% open



Climb crossings per km²

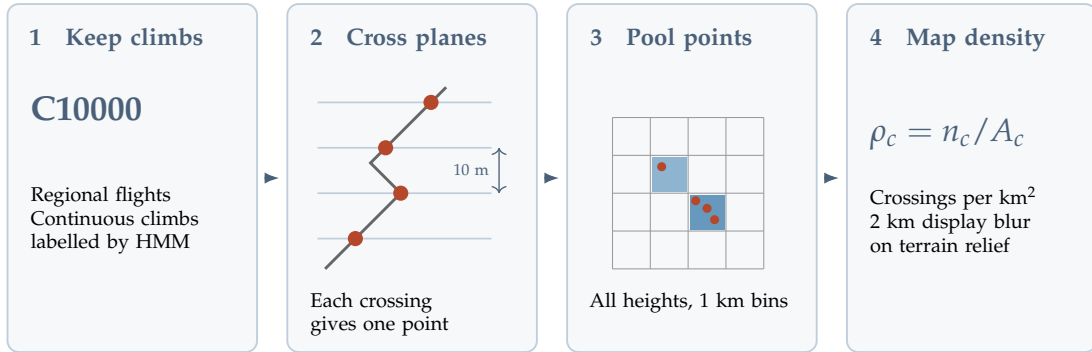


Both windows contain broad patches of climb use, with no single mountain-like crest corridor.

Their similar H values are compatible with weaker spatial constraints in lowlands.

C10000 paragliders; crossings every 10 m, all dates pooled. Matched 100 × 65 km lowland views; shared colour range. Map: © OSM contributors, SRTM, OpenTopoMap (CC-BY-SA).

The viewer's method: intersect continuous climbs with planes every 10 m, then pool the crossings.



One flight contributes at every crossing: a 500 m ascent gives about 50 points.

A: 4 altitudes

C: 2 circuits

E: beginners / experts

$4 \times 2 \times 2 = 16$ cells

$$H_{ace} = \underbrace{\mu + A_a + C_c + E_e}_{\text{additive part}}$$

$$+ (AC)_{ac} + (AE)_{ae} + (CE)_{ce} + (ACE)_{ace}$$

Each term contains one or more coefficients. Below: two altitude levels and two circuits, with wing group fixed.

Coefficients (sum-to-zero)

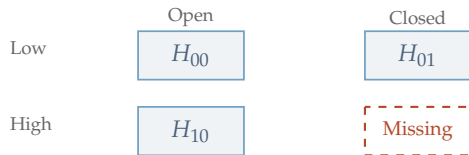
$\mu : 1, A : 3, C : 1, E : 1$

$AC : 3, AE : 3$

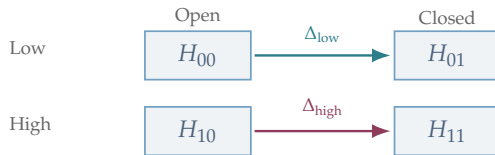
$CE : 1, ACE : 3$

$6 + 10 = 16$ cell means

OFAT: one combination missing



Full factorial: all four combinations



Interaction: $D_{AC} = \Delta_{\text{high}} - \Delta_{\text{low}} \neq 0$ means the circuit effect depends on altitude

What a group label requires

Flights in the same small cell, close enough in time to see each other: visual contact, about 15 minutes apart at most.

Then compare transport within shared altitude, circuit and wing-class settings.

$$\Delta H_{a,c,e}^{\text{social}} = H_{a,c,e,\text{group}} - H_{a,c,e,\text{solo}}$$

What the archive gives

Best case: the most populated 5 km cell (Low mountains), busiest day, peak 15 min.

≤ 63 flights

22 July 2023, 13:15–13:30 Paris time.
An upper bound, before any split over 32 cells. Other cells are sparser.

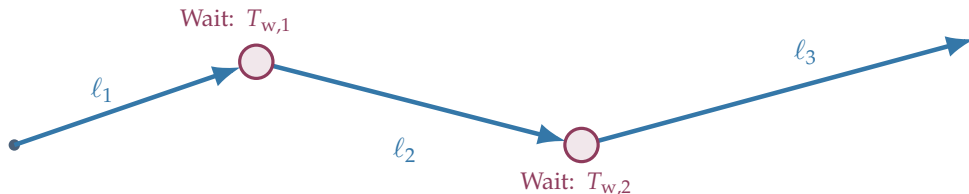
With the current archive, a solo versus group comparison is most likely out of reach

Displacement: gliding

Duration T_g , length ℓ and direction θ .
Retain turning-angle memory between glides.

Waiting: search + thermalling

T_w : total time between successive glides.
Estimate one pooled waiting-time distribution.



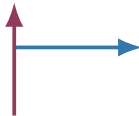
Waiting neglects local search motion and thermal drift.

Monte Carlo: test MSD, displacement distributions, anisotropy and memory

$$\mathbf{v}(t) = s(t) \mathbf{e}(t), \quad \mathbf{e} = (\cos \theta, \sin \theta) \quad \Delta\theta = \theta(t + \tau) - \theta(t)$$



Aligned: $\cos \Delta\theta = 1$



Perpendicular: 0



Opposite: -1

$$C_v(\tau) = \frac{\langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle}{\langle s^2 \rangle} = \frac{\langle s(t)s(t + \tau) \cos \Delta\theta \rangle}{\langle s^2 \rangle}$$

$$C_\theta(\tau) = \langle \mathbf{e}(t) \cdot \mathbf{e}(t + \tau) \rangle = \langle \cos \Delta\theta \rangle$$

Constant speed: $C_v = C_\theta$.

Persistent mean drift can leave a nonzero baseline; also inspect centred velocity covariance.

C_v combines speed and direction; C_θ isolates directional agreement

1 Group flights

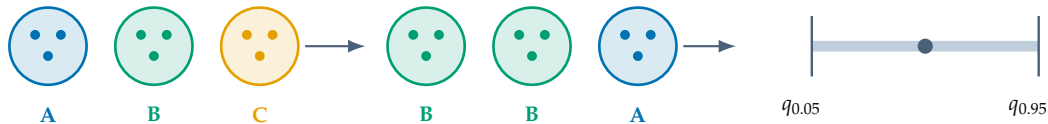
Take-off site $\times$ calendar day.
Aim: independent groups.

2 Resample groups

Sample with replacement.
Keep each group together.

3 Recompute

1,000 replicates.
Take the 5th–95th percentiles.



Current clusters are too small to absorb all shared flight conditions.

Intervals likely underestimate uncertainty; clustering is still being refined

Flight archive

FFVL Coupe Fédérale de Distance: IGC tracks, dates, declared sites, circuits and wing classes.

parapente.ffvl.fr/cfd/liste delta.ffvl.fr/cfd/liste

Maps and terrain

[OpenTopoMap](#): regional relief and contours (OSM / SRTM, CC-BY-SA).

[IGN / Géoplateforme](#): local Plan IGN backgrounds.

[IGN RGE ALTI](#): local terrain and derived crests.

Locator and analysis

[Natural Earth](#): France outline, public domain.

MSD, intervals and crossings: this thesis's prepared pipeline.

[Code, figure scripts and recorded source provenance](#)

Vilpellet, Darmon & Benzaquen (2026). *From Random Walks to Thermal Rides: Universal Anomalous Transport in Soaring Flights*. [arXiv:2601.01293](https://arxiv.org/abs/2601.01293).

Transport, HMM phase segmentation and phase-duration benchmarks.

NIST/SEMATECH. *e-Handbook of Statistical Methods*,
[§5.2.1.2: One Variable at a Time](#); [§5.3.3.3: Full Factorial Designs](#).

Mohr, Wilson & Freund (2022). *Statistical Methods*, 4th ed., §9.3, §9.7.

FAA (2024). *Glider Flying Handbook*, Ch. 9–10.

Soaring Society of America. *Lift Sources*.

Schroeder & Buck (1970). *Fire Weather*, Ch. 6: General Winds.

Savitzky & Golay (1964). Smoothing and Differentiation of Data by Simplified Least Squares Procedures. *Analytical Chemistry* 36, 1627–1639. [doi:10.1021/ac60214a047](https://doi.org/10.1021/ac60214a047).